Healthcare Data Visualization Using Python

Project Overview

Healthcare organisations generate large volumes of patient data every day. Proper analysis and visualization of this information enable healthcare professionals to identify trends, improve operational efficiency, and support evidence-based decision-making.

This project demonstrates how Python can be used to clean, analyse, and visualise healthcare data. Various exploratory data analysis (EDA) techniques were performed to understand patient demographics, medical conditions, billing information, hospital admissions, and insurance providers.

The project showcases practical data visualization skills using Pandas, Matplotlib, and Seaborn.

Project Objectives

The objectives of this project are to:

- Explore the healthcare dataset.
- Perform data cleaning and preprocessing.
- Analyse patient demographics.
- Examine relationships between healthcare variables.
- Produce meaningful visualisations.
- Generate business insights from healthcare data.

Dataset Description

The dataset contains patient healthcare records collected from a hospital information system. It includes demographic information, medical conditions, billing amounts, admission types, insurance providers, and hospital details.
The analysis began by inspecting the dataset structure and understanding the available variables before performing exploratory analysis.

```
df.head()
```

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
0	Bobby Jackson	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons And Miller	Blue Cross	18856.28	328	Urgent	2024-02-02	Paracetamol	Normal
1	Leslie Terry	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc	Medicare	33643.33	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
2	Danny Smith	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook Plc	Aetna	27955.10	205	Emergency	2022-10-07	Aspirin	Normal
3	Andrew Watts	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers And Vang,	Medicare	37909.78	450	Elective	2020-12-18	Ibuprofen	Abnormal
4	Adrienne Bell	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White	Aetna	14238.32	458	Urgent	2022-10-09	Penicillin	Abnormal

Figure 1: Dataset Preview

The dataset preview provides an overview of the available variables and confirms that each record represents a unique patient observation.

```
df.info()
<class 'pandas.core.frame.DataFrame'>
Index: 54966 entries, 0 to 55499
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Name                54966 non-null  object
 1   Age                 54966 non-null  int64
 2   Gender              54966 non-null  object
 3   Blood Type          54966 non-null  object
 4   Medical Condition   54966 non-null  object
 5   Date of Admission   54966 non-null  datetime64[ns]
 6   Doctor              54966 non-null  object
 7   Hospital            54966 non-null  object
 8   Insurance Provider  54966 non-null  object
 9   Billing Amount      54966 non-null  float64
 10  Room Number         54966 non-null  int64
 11  Admission Type      54966 non-null  object
 12  Discharge Date      54966 non-null  datetime64[ns]
 13  Medication          54966 non-null  object
 14  Test Results        54966 non-null  object
dtypes: datetime64[ns](2), float64(1), int64(2), object(10)
memory usage: 6.7+ MB
```

Figure 2: Dataset Information

The dataset information shows the number of observations, available variables, data types, and non-null values. This inspection confirmed that the dataset was suitable for further analysis.

```
df.dtypes
```

```
Name                          object
Age                            int64
Gender                        object
Blood Type                    object
Medical Condition             object
Date of Admission     datetime64[ns]
Doctor                        object
Hospital                      object
Insurance Provider            object
Billing Amount               float64
Room Number                    int64
Admission Type                object
Discharge Date        datetime64[ns]
Medication                    object
Test Results                  object
dtype: object
```

Figure 3: Variable Datatypes

Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Cleaning and Preparation

Before visualization, the dataset was carefully examined to improve data quality.
The preprocessing stage included:

- Checking missing values
- Removing duplicate records
- Verifying data types
- Identifying outliers
- Formatting variables appropriately

These steps ensured the dataset was reliable for subsequent analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset and identify important relationships among variables.
The analysis focused on:

- Descriptive statistics
- Correlation analysis
- Patient demographics
- Medical conditions
- Billing trends
- Hospital admissions

```
df.describe().T
```

	count	mean	min	25%	50%	75%	max	std
Age	54860.0	51.53385	13.0	35.0	52.0	68.0	89.0	19.605295
Date of Admission	54860	2021-11-01 19:01:50.506744576	2019-05-08 00:00:00	2020-07-28 00:00:00	2021-11-02 00:00:00	2023-02-03 00:00:00	2024-05-07 00:00:00	NaN
Billing Amount	54860.0	25594.633612	9.24	13299.75	25593.875	37847.065	52764.28	14175.867046
Room Number	54860.0	301.109752	101.0	202.0	302.0	400.0	500.0	115.217195
Discharge Date	54860	2021-11-17 07:00:08.137076224	2019-05-09 00:00:00	2020-08-13 00:00:00	2021-11-18 00:00:00	2023-02-19 00:00:00	2024-06-06 00:00:00	NaN

Figure 4: Descriptive Statistics

The descriptive statistics summarize the central tendency and variability of numerical variables within the dataset. These statistics provide an initial understanding of patient characteristics.

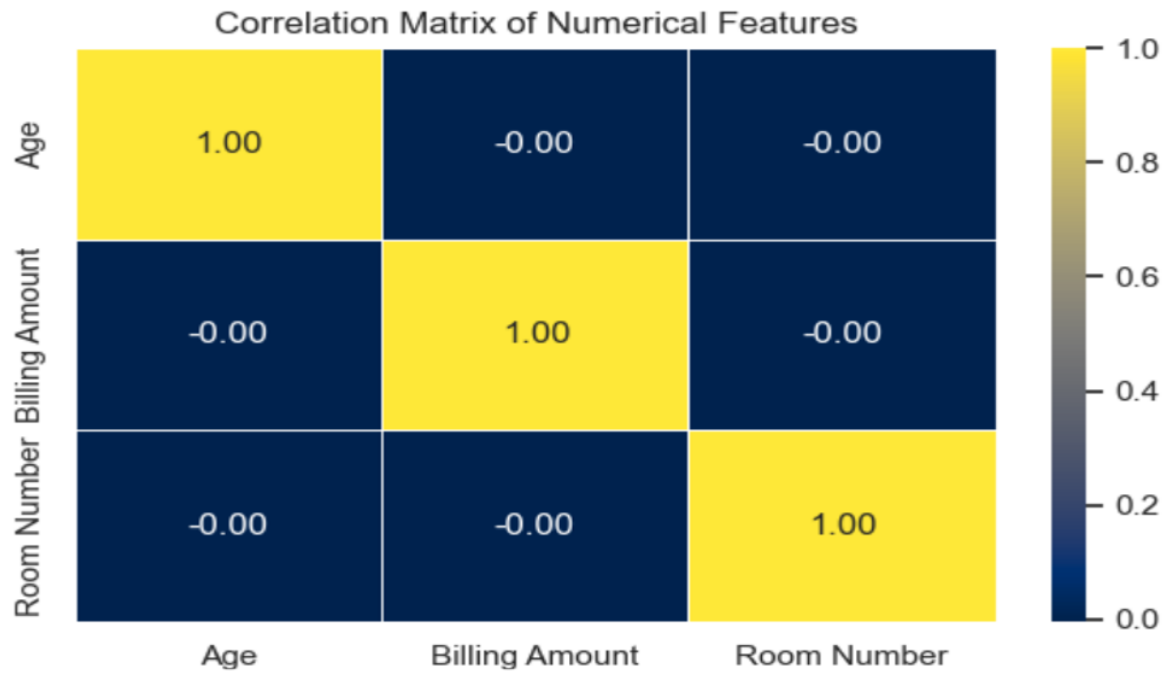


Figure 5: Correlation Heatmap

The correlation heatmap illustrates the relationships among numerical variables. Most variables exhibit weak correlations, indicating that age, billing amount, and room number are largely independent of one another.

Data Visualization

Several visualisations were produced to explore different aspects of the healthcare dataset.

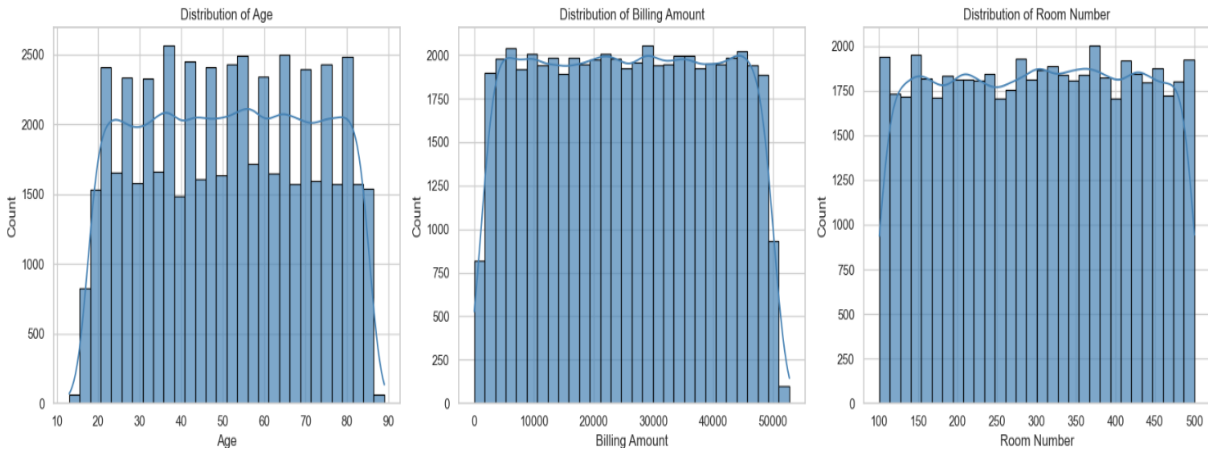


Figure 6: Distribution of Numerical Variables

The figure above presents the distribution of three important numerical variables: **Age**, **Billing Amount**, and **Room Number**.

- **Age:** The age distribution shows that most patients are between **20 and 80 years** old, with a higher concentration among middle-aged and older adults. The distribution is

relatively balanced, indicating that the dataset contains patients from a wide range of adult age groups.

- **Billing Amount:** Billing amounts are distributed across a broad range, with most patient charges falling between **10,000 and 50,000** monetary units. The distribution appears even, although slightly fewer observations occur at the extreme lower and upper ends of the range.
- **Room Number:** Room numbers are evenly distributed between approximately **100 and 500**, suggesting that patient admissions were spread across different hospital rooms without any obvious concentration in a particular section of the hospital.

Key Insights:

The distributions indicate that the dataset captures a diverse patient population with varying healthcare costs and hospital room allocations. None of the three variables exhibit severe imbalance, making them suitable for subsequent exploratory data analysis and visualization.

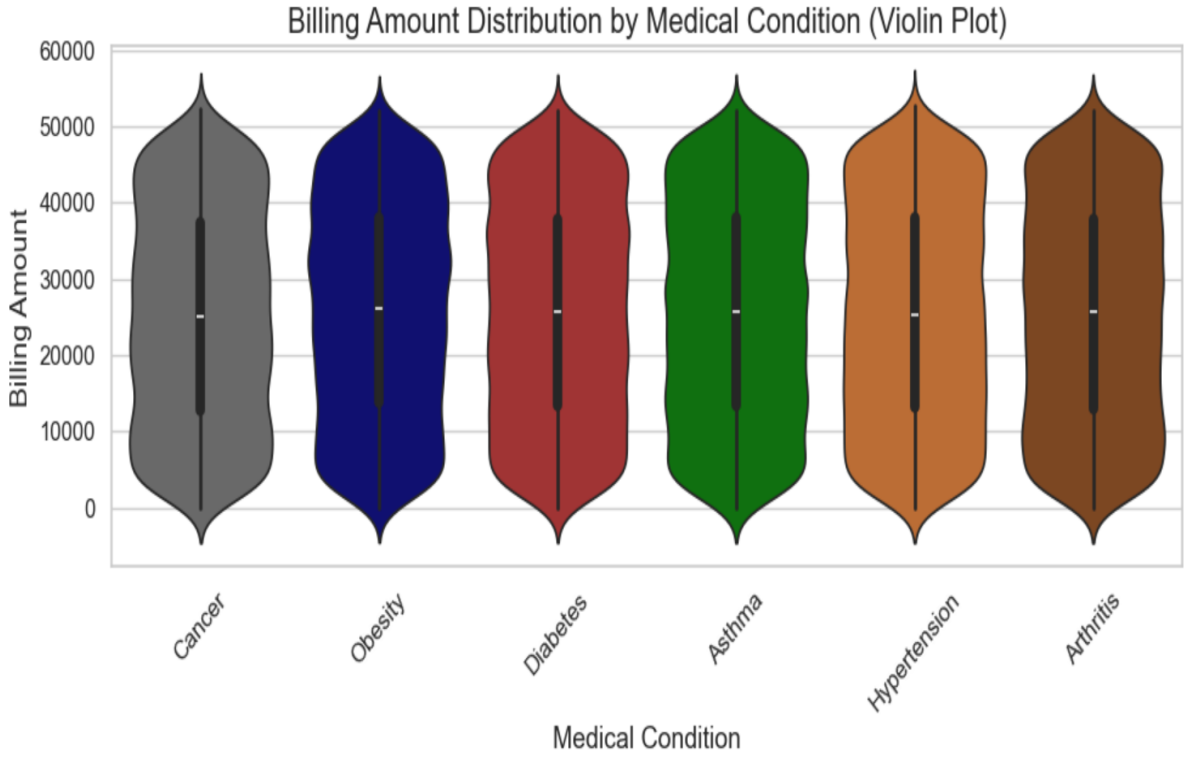


Figure 7: Billing Amount by Medical Condition

The violin plot illustrates the distribution of billing amounts across different medical conditions. While billing distributions are generally similar across conditions, slight variations suggest that certain conditions may incur higher treatment costs.

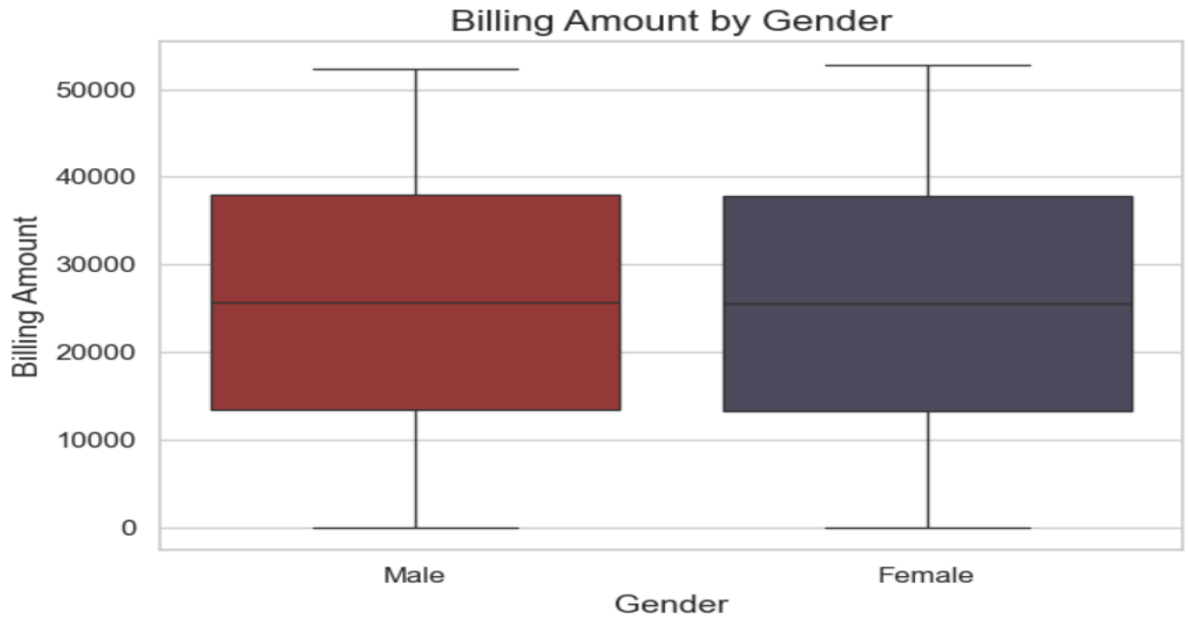


Figure 8: Box plot of Billing Amount by Gender

The boxplot indicates that male and female patients have comparable billing distributions, suggesting that healthcare costs are relatively consistent across gender.

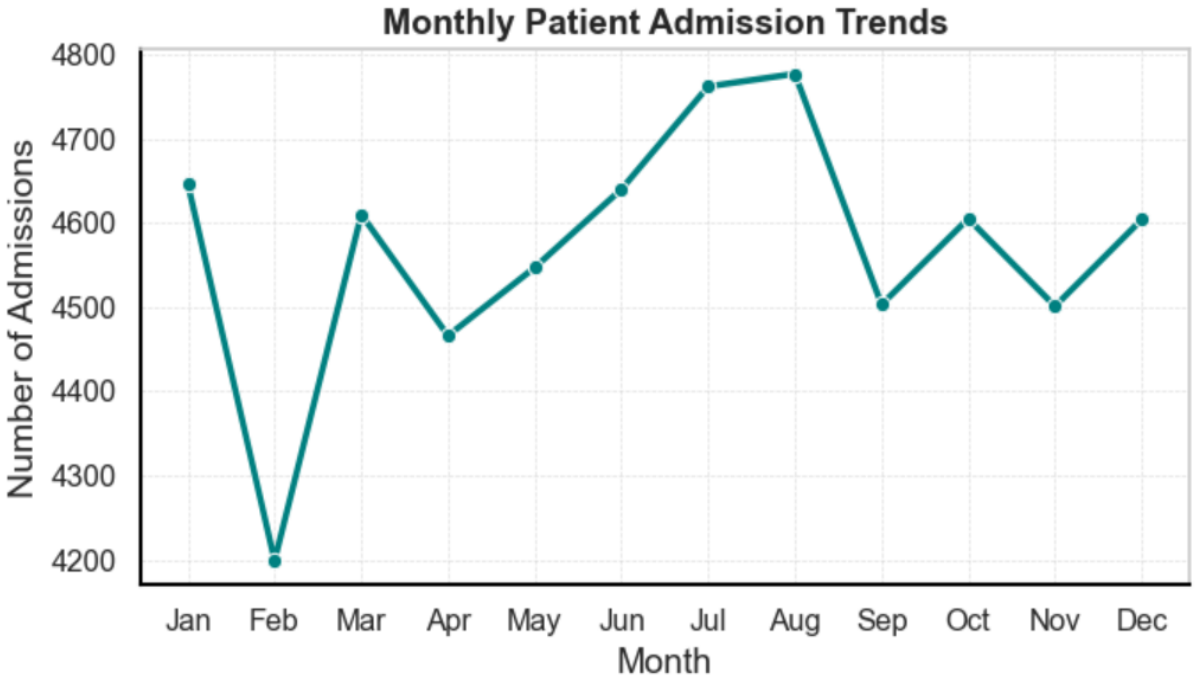


Figure 9: Patients Admissions Trend Over Different Months

Patient admissions varied throughout the year, with the highest number recorded in August at 4,832 and the lowest in February at 4,255, indicating a possible seasonal pattern. These fluctuations highlight the importance of aligning staffing, service preparation, and resource allocation with expected changes in patient demand.

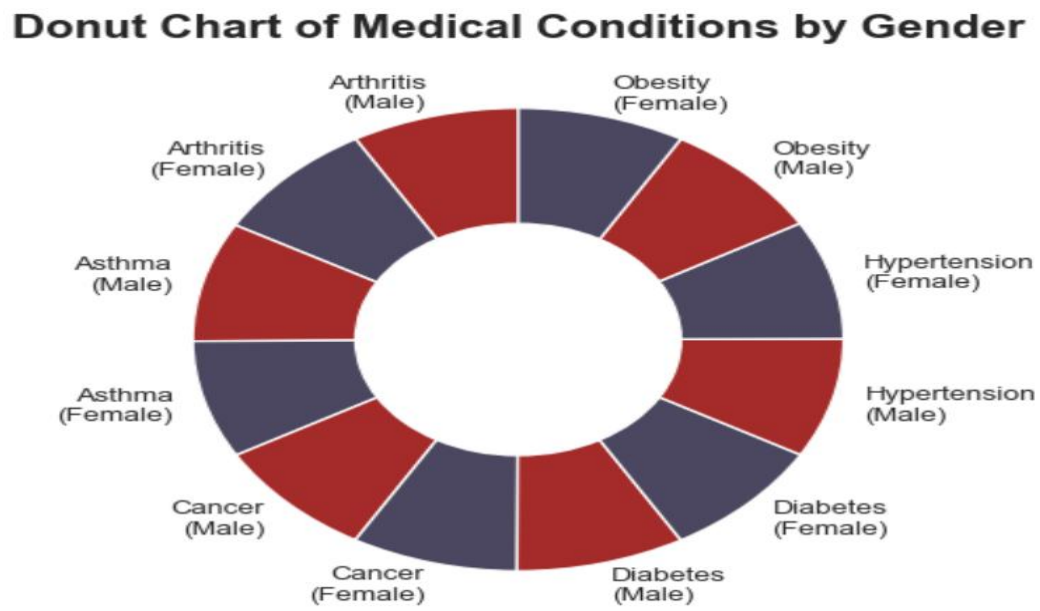


Figure 10: Donut Chat of Medical Condition by Gender

The chart shows a generally balanced gender distribution across most medical conditions, although certain diseases, such as hypertension and diabetes, appear to be more prevalent in one gender. These findings suggest possible gender-related differences in disease occurrence and provide useful insights for developing targeted healthcare interventions and treatment strategies.

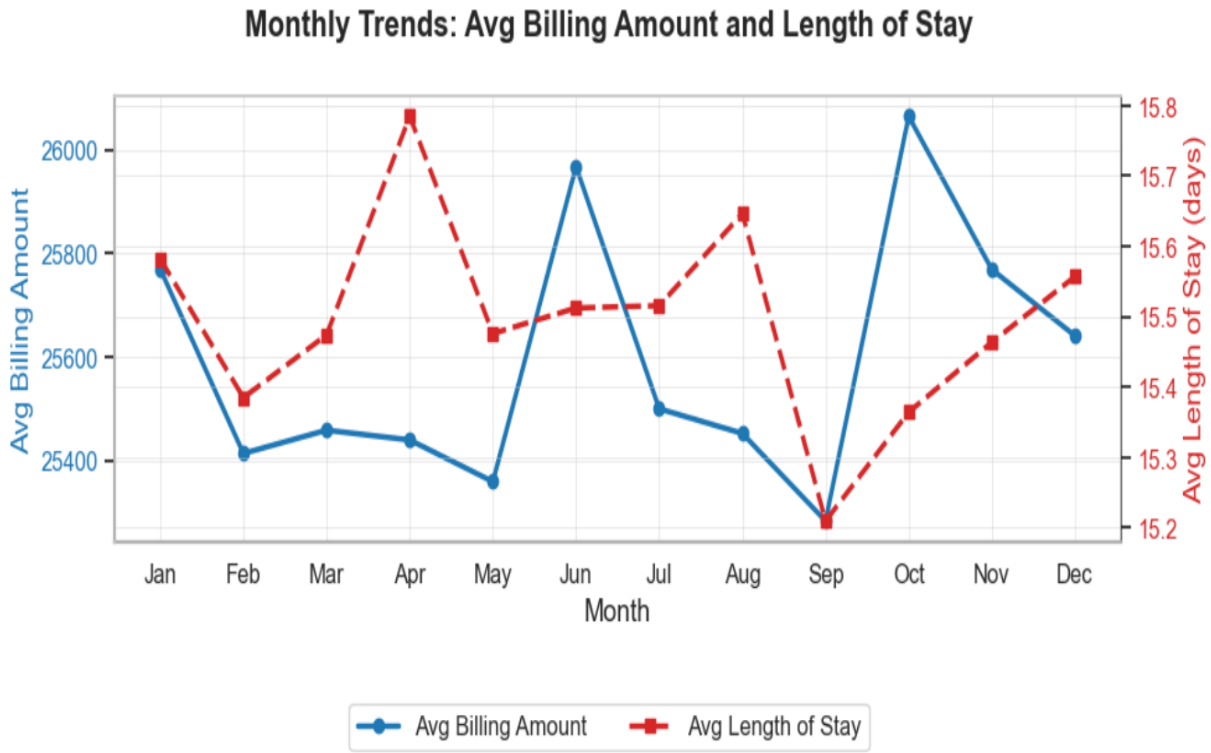


Figure 11: Average Billing Amount and the Length of Stay

October recorded the highest average billing, while April had the longest average hospital stay, suggesting that higher healthcare costs are not always associated with longer hospitalizations. These findings provide valuable insights for improving budget planning, staffing strategies, and resource allocation within healthcare facilities.

Key Findings

The analysis revealed several important observations:

- The dataset contained a diverse patient population across different age groups.
- Billing amounts varied considerably among patients.
- Hospital admissions showed seasonal fluctuations throughout the year.
- Billing costs differed across medical conditions.
- Gender distribution was relatively balanced across most medical conditions.
- Data visualization successfully revealed patterns that support healthcare planning and operational decision-making.

Recommendations

Based on the findings, the following recommendations are proposed:

- Improve data quality through regular validation procedures.
- Monitor patient demographics continuously to identify healthcare trends.
- Use dashboards for real-time healthcare monitoring.
- Expand the dataset with additional clinical variables.
- Integrate predictive analytics to support clinical decision-making.
- Develop an interactive dashboard using Power BI or Tableau for real-time healthcare monitoring. That shows employers you understand business intelligence tools.

Conclusion

This project demonstrates how Python can be effectively used for healthcare data exploration and visualization. Through data cleaning, exploratory analysis, and visual storytelling, meaningful insights were generated from patient healthcare records.

The project highlights the practical application of data analytics techniques within the healthcare sector and provides a strong foundation for future predictive modelling and dashboard development. This project demonstrates practical skills in data cleaning, exploratory data analysis, and data visualization using Python, making it suitable for portfolio presentation and future healthcare analytics projects.